

Total No. of Questions : 6]

P616

SEAT No. :

[Total No. of Pages : 2

BE/Insem/APR - 249
B.E. (Computer Engineering)
EMBEDDED AND REAL TIME OPERATING SYSTEM
(2015 Pattern) (Semester - II) (Elective - III)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Solve Q.1 or 2, Q.3 or 4, Q.5 or 6.*
- 2) Neat diagrams must be drawn wherever necessary*
- 3) Figures to the right indicate full marks.*
- 4) Assume suitable data, if necessary.*

Q1) a) List and Explain software tools used for designing of an embedded system. **[5]**

b) With neat diagram, explain Embedded hardware units and devices in a system. **[5]**

OR

Q2) a) What are different characteristics and challenges of embedded system? **[5]**

b) List and explain various types of embedded system processors. **[5]**

Q3) a) What the main characteristics of CAN bus standards? Draw and explain the CAN data frame format. **[6]**

b) What is Embedded System on Chip (SoC). Explain all components embedded in SoC. **[4]**

OR

Q4) a) Explain in detail ARM Processor Architecture with core architectural block diagram. **[6]**

b) Enlist various Wireless and mobile system Protocols and explain any one with its features and diagram. **[4]**

P.T.O.

Q5) a) What is HDLC? Explain different frame types and data transfer modes of HDLC. [5]

b) Compare ISA and PCI buses based on their features such as data rate, data size, connector size and applications. [5]

OR

Q6) a) Differentiate : [5]

i) Serial and Parallel communication.

ii) Synchronous and asynchronous communication.

b) Draw and Explain Parallel ports and its interfacing with the stepper motor. [5]

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